

Ananya Appan

RESEARCH INTERESTS

Cryptography and Distributed Computing: Secure Multi Party Computation, Verifiable Secret Sharing, Consensus, Oblivious RAM

EDUCATION

Ph.D (2nd year)

Aug 2023 - Present

University of Illinois Urbana Champaign,

CGPA : 4 / 4

Supervisors : David Heath, Ling Ren

Integrated Master of Technology

July 2022

International Institute of Information Technology (IIIT) Bangalore, India

CGPA : 3.84 / 4

Supervisor : Ashish Choudhury

Intermediate (12th Class)

May 2017

Chetana Pre University College, Bangalore, India

Karnataka Pre University Board

Percentage : 98%

Secondary (10th Class)

March 2015

Delhi Public School Bangalore North

CBSE (Central Board of Secondary Education)

CGPA : 10 / 10

JOURNAL PUBLICATIONS

Ananya Appan, Anirudh Chandramouli, Ashish Choudhury, Perfectly-Secure Synchronous MPC with Asynchronous Fallback Guarantees, *IEEE Transactions on Information Theory*, 2023

Ananya Appan, Anirudh Chandramouli, Ashish Choudhury, Revisiting the Efficiency of Asynchronous MPC with Optimal Resilience Against General Adversaries, *Journal of Cryptology*, 2023

CONFERENCE PUBLICATIONS

Ananya Appan, David Heath, Ling Ren Verifiable Secret Sharing, Randomized Agreement, and Multi-Party Computation in Granular Synchrony : *ASIACRYPT* 2025

Ananya Appan, David Heath, Ling Ren Oblivious Single Access Machines: A New Model for Oblivious Computing : *ACM Conference on Computer and Communications Security (CCS)* 2024

Ananya Appan, Ashish Choudhury, Network Agnostic MPC with Statistical Security : *Theory of Cryptology Conference (TCC)* 2023

Ananya Appan, Ashish Choudhury, Perfectly Secure Synchronous MPC with Asynchronous Fallback Guarantees Against General Adversaries : *International Symposium on Distributed Computing (DISC)* 2023

Ananya Appan, Anirudh Chandramouli, Ashish Choudhury, Perfectly-Secure Synchronous MPC with Asynchronous Fallback Guarantees : *ACM Symposium on Principles of Distributed Computing (PODC)* 2022

- Ananya Appan, Anirudh Chandramouli, Ashish Choudhury, [Revisiting the Efficiency of Asynchronous Multi Party Computation Against General Adversaries](#) : *INDOCRYPT 2022: 23rd International Conference on Cryptology*

UNPUBLISHED MANUSCRIPTS

- Ananya Appan, David Heath: Oblivious Immutable Memory

TALKS

Invited talk at [University of Connecticut](#) Security Seminar Nov 2024

- Based on the paper “Oblivious Single Access Machines”

[Bangalore Crypto Day, Spring 2023](#) March 2023

- Based on the paper “Revisiting the Efficiency of Asynchronous Multi Party Computation Against General Adversaries” at the Indian Institute of Science.

[Theory and Practice of Multi Party Computation \(TPMPC\) 2022](#) June 2022

- Remote talk based on the paper “Perfectly-Secure Synchronous MPC with Asynchronous Fallback Guarantees” ([recording](#)).

TEACHING ASSISTANT- SHIPS

International Institute of Information Technology, Bangalore

Conducted (online) tutorial sessions, set questions for assignments and exams, and evaluated these for the following courses.

- Foundations of Cryptography January 2022 - May 2022.
Instructors : Dr. [Ashish Choudhury](#), Dr. [Srinivas Vivek](#)
- Discrete Mathematics August 2021 - December 2021.
Instructor : Dr. [Ashish Choudhury](#)
- Data Structures and Algorithms March 2021 - July 2021.
Instructor : Dr. [V N Muralidhara](#)
- Programming II (C++ and Java) August 2020 - December 2020.
Instructors : Dr. [Jaya Sreevalsan Nair](#), Dr. [T K Srikanth](#)

[National Programme on Technology Enhanced Learning \(NPTEL\)](#)

Instructor : Dr. [Ashish Choudhury](#)

Set questions for assignments and exams, and helped manage the discussion forum for the following MOOC (Massive Open Online Courses) courses.

- [Secure Computation: Part II](#) July 2022 - November 2022.
- [Discrete Mathematics](#) January 2022 - April 2022.
- [Foundations of Cryptography](#) January 2022 - April 2022.
- [Secure Computation: Part I](#) July 2021 - November 2021.

WORK EXPERIENCE

Associate Developer July 2022 - Present
[SAP Labs, India](#)

- Build cloud applications on SAP’s Business Technology Platform ([BTP](#)) using SAP Cloud Application Programming ([CAP](#)) model.

Summer Intern May 2021 - July 2021
[SAP Labs, India](#)

- Created dashboards using SAP Analytics Cloud ([SAC](#)) and worked on a proof of concept for querying encrypted databases.

Research Intern May 2020 - July 2020
[Indian Institute of Science, Bangalore](#)

- Supervisor : Dr. [Jayant R. Haritsa](#)

- Compared [UNMASQUE](#), a hidden SQL query extractor, to other implementations for reverse query engineering, and built a synthetic SQL query generator for testing it.

Developer

June 2018 - December 2018

LexHeal, Bangalore

- Helped build the front end of an EHR Mobile Application for a start up using React Native.

ACHIEVEMENTS

- Received the Late Sri. N. Rama Rao Gold Medal for [Student of the Year](#) for the graduating batch of 2022.
- Included in the [Dean's Merit List](#) of IIIT Bangalore from 2017 to 2022.
- Selected for being awarded the [IAS](#) (Indian Academy of Sciences) fellowship under the Computer Science Engineering category in 2020.
- Finalist in [LinkedIn Wintathon 2020](#).
- Part of the team which received second place in the [Codess](#) Hackathon conducted by Microsoft in 2019.
- Qualified for [ACM ICPC](#) Regionals (Amritapuri and Kharagpur) in 2018.
- Among top 1000 students in the country selected for being awarded the [KV PY](#) Scholarship in 2016.
- Ranked 24th in the state in the NTSE (National Talent Search Exam) in 2014.

PROJECTS

Autism Detection In Children

Project elective done in collaboration with St. Johns Centre for Advanced Research and Excellence in Autism and Developmental Disorders ([CAREADD](#)).

- Supervisors : Dr. [Dinesh Babu Jayagopi](#), Dr. [Shyam Sundar Rajagopalan](#)
- Fine tuned a 3D convolutional neural network pretrained on the Kinetics dataset to detect stereotypical behaviour in autistic children. Tried to use self supervised learning to augment the data used to train the network.

Web App to create DocBooks online

- Supervisor : Dr. [Chandrashekar Ramanathan](#)
- Implemented a web application to perform CRUD operations on a book written in the DocBook XML format. Focused on how content rendered in DocBook XML can be ported to other data formats like PDF and HTML.

Saliency Deblurring

- Supervisor : Dr. [Dinesh Babu Jayagopi](#)
- Trained an encoder-decoder network to deblur images based on saliency maps. Used a multi-head decoder architecture to deblur foreground and background separately.